

VMware ESXi

PHASE 1 - PROFESSIONAL STUDY GUIDE

Roman Urdu + English Technical Terms

Virtualization • Hypervisors • ESXi • Management • vSphere Standard Networking

Maqsad: Concepts ko asaan Roman Urdu mein samajhna aur technical terms ko English mein yaad rakhna.

Use: Class revision, lab preparation aur interview practice.

SAB SE IMPORTANT NETWORK FLOW

VM → vNIC → VM Port Group → vSwitch → vmnic → Physical Network

Phase 1 Complete Study Pack

1. VMware Quick Revision - Roman Urdu

1 Virtualization

Ek physical system ke resources use karke multiple VMs run karna.

Yaad rakhein: Physical Server → Hypervisor → Multiple VMs

2 Hypervisor

Software/platform jo CPU, RAM, storage aur networking resources ko VMs ke liye manage aur allocate karta hai.

Yaad rakhein: Hardware → Hypervisor → VMs

3 Type 1 vs Type 2

Type 1 hardware par directly run hota hai. Type 2 host OS ke upar run hota hai.

Yaad rakhein: ESXi = Type 1 | Workstation = Type 2

4 VMware / ESXi

VMware virtualization industry ka important vendor raha hai aur ab Broadcom ka hissa hai. Hamara focus ESXi hai.

Yaad rakhein: ESXi = VMware Type 1 Hypervisor

5 ESX vs ESXi

ESX older architecture tha jisme traditional Service Console thi. ESXi newer purpose-built architecture hai.

Yaad rakhein: ESX = old | ESXi = modern family

6 DCUI

Direct Console User Interface. Local/basic ESXi configuration aur troubleshooting ke liye.

Yaad rakhein: IP, subnet, gateway, DNS, hostname

7 vSwitch

ESXi host ke andar software-based Layer 2 virtual switch.

Yaad rakhein: VMs → Port Group → vSwitch → vmnic

8 vNIC vs vmnic

vNIC VM ka virtual network adapter hai. vmnic ESXi host ki physical NIC/uplink ko represent karta hai.

Yaad rakhein: vNIC = virtual | vmnic = physical uplink

9 VM Port Group

VMs ke traffic, VLAN configuration aur common network policies ke liye.

Yaad rakhein: VM → vNIC → VM Port Group

10 VMkernel Port

ESXi host ke apne network/services ke liye, jaise management aur vMotion.

Yaad rakhein: Host/service traffic → VMkernel Port

11 NIC Teaming

Multiple physical uplinks redundancy/failover aur policy-based traffic distribution ke liye.

Yaad rakhein: vSwitch → vmnic0 + vmnic1

12 Traffic Shaping

Network traffic ki bandwidth/rate ko policy ke mutabiq control karta hai.

Yaad rakhein: Average | Peak | Burst

2. Study Roadmap

SECTION	KYA SEEKHNA HAI
A. Virtualization Foundation	Virtualization, VM, Host vs Guest, Hypervisor, Type 1 vs Type 2
B. ESXi Architecture & Management	ESX vs ESXi, DCUI, VMkernel aur browser-based management
C. ESXi Networking	vSS, vNIC, vmnic, VM Port Group, VMkernel Port, NIC Teaming, Traffic Shaping
D. Interview Practice	20 important Phase 1 questions aur interview-friendly answers
E. Final Memory Sheet	Quick recall aur VM se physical network tak complete traffic path

Study karne ka best tareeqa

1	Quick Revision	Pehle concepts ko 2-3 minute mein revise karein.
2	Detailed Concepts	Har term ka purpose samjhein, sirf definition yaad na karein.
3	Network Flow	VM → vNIC → Port Group → vSwitch → vmnic flow bol kar practice karein.
4	Interview Q&A	Pehle khud jawab dein, phir model answer check karein.

3. Virtualization Foundations

Virtualization

Virtualization ek technology hai jis se hum ek physical server ke CPU, RAM, storage aur networking resources ko use karke multiple Virtual Machines chala sakte hain.

Easy memory: One physical machine → Multiple VMs

Virtual Machine (VM)

VM ek software-based computer hai. Iske paas vCPU, RAM, Virtual Disk, vNIC, Operating System aur applications ho sakte hain.

Easy memory: VM ke resources physical host se aate hain.

Host vs Guest

Host woh system hai jo virtualization resources provide karta hai. Guest woh operating system hai jo VM ke andar run karta hai.

Easy memory: Windows Host → Workstation → Rocky Linux Guest

Hypervisor

Hypervisor physical hardware resources ko manage karta hai aur VMs ko allocate karta hai.

Easy memory: CPU / RAM / Disk / NIC → Hypervisor → VMs

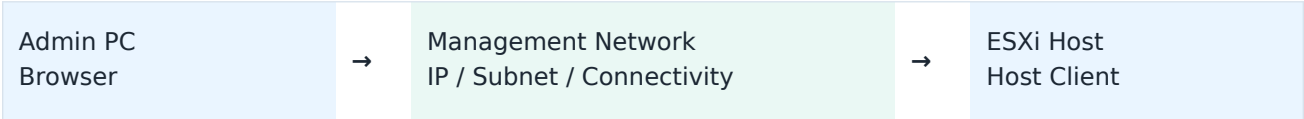
Type 1 vs Type 2 Hypervisor

	TYPE 1 - BARE-METAL	TYPE 2 - HOSTED
Kahan run hota hai	Physical hardware par directly	Host operating system ke upar
Example	VMware ESXi	VMware Workstation
Common use	Servers / data centers	Desktop / labs
Flow	Hardware → ESXi → VMs	Hardware → Windows/Linux → Workstation → VMs

4. ESXi Architecture & Management

TOPIC	ROMAN URDU EXPLANATION
ESX vs ESXi	ESX older VMware architecture tha jisme traditional Linux-based Service Console thi. ESXi newer p
DCUI	Direct Console User Interface. ESXi host ke local console se management network, IP, subnet, gate
VMkernel	ESXi ka core component hai jo CPU scheduling, memory, storage, networking aur VMs ke hardware
ESXi Host Client	Standalone ESXi host ko browser se management IP ke through manage kiya ja sakta hai.

Management Connectivity



5. ESXi Networking

Phase 1 ka sab se important networking flow

1	VM	Guest workload / virtual machine
2	vNIC	VM ke andar virtual network adapter
3	VM Port Group	VM connectivity aur common network policies
4	vSwitch	ESXi ka software-based Layer 2 virtual switch
5	vmnic	ESXi host ki physical NIC/uplink representation
6	Physical Switch	External physical network switch
7	Physical Network	LAN / routed network, design ke mutabiq

Golden Rule

VM ko network chahiye
VM PORT GROUP

ESXi host/service ko network chahiye
VMKERNEL PORT

TERM	PURPOSE
vSphere Standard Switch (vSS)	ESXi host ke andar software-based Layer 2 virtual switch.
vNIC	Virtual Machine ka virtual network adapter.
vmnic	ESXi host ki physical network adapter/uplink ka naam/representation.
VM Port Group	VM traffic, VLAN configuration aur common network policies.
VMkernel Port	ESXi host/service traffic - management, vMotion aur configuration ke mutabiq certain storage
NIC Teaming	Multiple uplinks redundancy/failover aur configured policy ke mutabiq traffic distribution ke liye
Traffic Shaping	Network traffic ki bandwidth/rate ko configured policy ke mutabiq control karta hai.

6. Interview & Self-Test - 20 Questions

Har question ka jawab pehle khud bolne ki koshish karein. Phir neech model answer check karein.

Q1	Virtualization kya hai?
	Virtualization ek technology hai jis se hum ek physical server ke CPU, RAM, storage aur networking resources ko use karke multiple Virtual Machines chala sakte hain.
Q2	VM kya hoti hai?
	VM yani Virtual Machine ek software-based computer hoti hai. Iske paas vCPU, RAM, virtual disk, vNIC aur operating system ho sakta hai, lekin underlying resources physical host se aate hain.
Q3	Host aur Guest mein kya difference hai?
	Host woh system hai jo virtualization resources provide karta hai. Guest woh operating system hai jo VM ke andar run karta hai.
Q4	Hypervisor kya karta hai?
	Hypervisor physical server ke CPU, RAM, storage aur networking resources ko manage karta hai aur VMs ko resources allocate karta hai.
Q5	Type 1 aur Type 2 Hypervisor mein kya difference hai?
	Type 1 physical hardware par directly run karta hai. Type 2 existing host operating system ke upar application ki tarah run karta hai.

Q6	ESXi Type 1 hai ya Type 2?
	VMware ESXi Type 1 (Bare-Metal) Hypervisor hai.
Q7	VMware Workstation Type 1 hai ya Type 2?
	VMware Workstation Type 2 Hypervisor hai kyun ke ye Windows ya Linux host operating system ke upar run karta hai.
Q8	ESX aur ESXi mein basic difference kya hai?
	ESX older VMware hypervisor architecture tha jisme traditional Linux-based Service Console thi. ESXi newer purpose-built architecture hai jisme traditional ESX Service Console nahi hoti.
Q9	DCUI kya hai?
	DCUI ka full form Direct Console User Interface hai. Ye ESXi host ki local console interface hai jahan se basic configuration aur troubleshooting ki ja sakti hai.
Q10	VMkernel kya hai?
	VMkernel ESXi ka core component hai. Ye CPU scheduling, memory management, storage, networking aur VMs ke hardware-resource access jaise important functions handle karta hai.

Q11 vSphere Standard Switch kya hai?

vSphere Standard Switch (vSS) ESXi host ke andar software-based Layer 2 virtual switch hai jo VMs aur VMkernel networking ko connectivity provide karta hai.

Q12 vNIC kya hai?

vNIC yani Virtual Network Interface Card, VM ka virtual network adapter hota hai.

Q13 vmnic0 kya represent karta hai?

vmnic0 ESXi host ki physical network adapter/uplink ko represent karta hai.

Q14 VM Port Group ka purpose kya hai?

VM Port Group VMs ko network connectivity aur common network configuration/policies provide karta hai. VLAN configuration bhi Port Group par ho sakti hai.

Q15 VMkernel Port ka purpose kya hai?

VMkernel Port ESXi host ke apne network traffic aur services ke liye use hota hai, jaise management, vMotion aur configuration ke mutabiq certain storage traffic.

Q16 VM Port Group aur VMkernel Port mein difference kya hai?

VM Port Group VM traffic ke liye hota hai, jabke VMkernel Port ESXi host/service traffic ke liye hota hai.

Q17 NIC Teaming kyun use karte hain?

NIC Teaming multiple physical uplinks use karke redundancy/failover provide kar sakti hai aur configured teaming policy ke mutabiq traffic distribution mein help kar sakti hai.

Q18 Traffic Shaping kya control karti hai?

Traffic Shaping network traffic ki bandwidth/rate ko configured policy ke mutabiq control karti hai. Important terms Average Bandwidth, Peak Bandwidth aur Burst Size hain.

Q19 ESXi host ko browser se access karne ke liye management networking kyun zaroori hai?

Admin PC aur ESXi host network ke through communicate karte hain. Isliye ESXi ko proper management IP, subnet configuration aur network connectivity chahiye. Routing ki zarurat ho to gateway important hota hai.

Q20 Ek VM ka traffic physical network tak kis path se ja sakta hai?

Typical path: VM → vNIC → VM Port Group → vSwitch → vmnic → Physical Switch → Physical Network.

7. Final Memory Sheet

TERM	YAAD RAKHEIN
ESXi	Type 1 Bare-Metal Hypervisor
VMware Workstation	Type 2 Hosted Hypervisor
VMkernel	ESXi ka core component
DCUI	Local/basic ESXi configuration
vSwitch	Virtual Layer 2 switch
vNIC	VM ka virtual NIC
vmnic	ESXi host ki physical NIC/uplink
VM Port Group	VM traffic
VMkernel Port	ESXi host/service traffic
NIC Teaming	Multiple uplinks / redundancy
Traffic Shaping	Bandwidth/rate control

Must-Memorize Traffic Path

VM → vNIC → VM Port Group → vSwitch → vmnic → Physical Switch → Network

Self-Test Checklist

☐ Virtualization	☐ VM
☐ Host vs Guest	☐ Hypervisor
☐ Type 1 vs Type 2	☐ ESX vs ESXi
☐ DCUI	☐ VMkernel
☐ vSphere Standard Switch	☐ vNIC
☐ vmnic	☐ VM Port Group
☐ VMkernel Port	☐ NIC Teaming
☐ Traffic Shaping	☐ Complete VM-to-physical-network flow